CRYPTOMNIA

The Memory Layer for Crypto

What Was True When

Proving what was known at the moment a decision was made —
for compliance, audit, research, and risk.

Built on Pod-OS by PointOfData

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Abstract

On-chain data is permanent. The meaning of that data is not. Risk scores are revised, entity attributions are updated, sanctions lists change, and audit conclusions are overturned. Most crypto analytics and compliance tools present only the current view of an address or transaction. This creates a structural gap: when a regulator, auditor, or investigator asks what was known at the time a decision was made, organizations cannot reliably answer.

Cryptomnia is the memory layer for crypto. It stores every risk score, attribution, and claim as a time-stamped Event Object with full provenance. Built on Pod-OS — a time-aware multigraph database — Cryptomnia preserves previous versions, makes forgetting auditable, and enables precise “as-of” queries. Supporting evidence (files, screenshots, reports) can be attached to any observation and retrieved later. The result is institutional memory that can prove what was true when.

1. The Problem: Permanent Data, Transient Meaning

Blockchains record transactions immutably. Once confirmed, the raw facts of who transferred what to whom do not change. Compliance, risk, and investigative work, however, depend on interpretive layers built on top of those facts: risk scores, entity labels, clustering, sanctions exposure, and qualitative assessments.

These interpretive layers evolve:

- A Chainalysis or TRM risk score of 92 (high mixer exposure) can later become 45 after internal review clears the address.
- An address initially attributed to a legitimate exchange may later be linked to a sanctioned entity or illicit service.
- A source that once informed a decision may be discredited; the original conclusion must be preserved even as it is superseded.
- Sanctions lists and regulatory guidance are updated; prior activity must be re-evaluated against the information available at the time.

Existing tools are optimized for the present. They surface the latest score, the current attribution, and the most recent alert. When asked what was known on a specific date, organizations must reconstruct the answer from logs, emails, and memory — or admit they cannot.

2. The Cryptomnia Approach

Cryptomnia treats every risk score, attribution, and claim as a first-class, time-stamped Event Object. Previous versions are never overwritten. When a source is discredited, the original assessment is marked forgotten; the forgetting event itself becomes a permanent, queryable record. Supporting evidence — screenshots, reports, internal notes, or external documents — can be attached as linked Event Objects and retrieved later with full provenance.

3. Built on Pod-OS

Cryptomnia is built on Pod-OS, the time-aware multigraph database developed by PointOfData. Pod-OS was designed to solve a broader problem: most AI and enterprise systems fail not because of models, but because facts are scattered, stripped of temporal context, and frozen on their last version.

In Pod-OS, every node is an Event Object — a single asserted claim that carries:

- Its embedding (for semantic similarity and retrieval)
- Typed fields
- A validity interval (when it was true)
- Its source and confidence
- Tags that describe it

Event Objects are connected by weighted, bi-directional links, forming a multigraph that can be traversed from either end. Nothing is overwritten: a change is a new assertion with its own time and source. Because facts are typed and timestamped at the claim level, the store can answer questions that conventional vector, graph, or relational systems cannot express cleanly — as-of state, what changed between two dates, contradictions between sources, and what is missing.

4. Data Model for Crypto Memory

Cryptomnia applies the Pod-OS Event Object model to the domain of crypto risk and attribution using an Observation / Entity pattern.

Entities

Addresses, clusters, services, or counterparties are represented as entities. Tags on these entities enable direct lookup (for example, by address).

Observations

Each risk score, attribution, or claim is an Observation stored as an Event Object. The Observation carries the score or label, the source (e.g., Chainalysis, internal review), the timestamp, and ownership linking it to the entity it describes. Successive observations for the same entity accumulate over time without overwriting prior ones.

Evidence

Supporting materials — files, screenshots, internal reports, or external documents — are stored as separate Evidence Event Objects and linked to the Observation they support. File content is retained (base64-encoded in tags) so it remains retrievable even if the original payload is later replaced by link metadata. Evidence carries its own timestamp, source, and description, preserving the chain of what was available when a decision was made.

Links and Provenance

Links connect observations to entities, to supporting evidence, and to related claims. When a source is discredited, a new “forgotten” Event Object records the withdrawal while the original remains addressable.

5. Worked Example: Successive Risk Scores and As-Of Queries

Consider an address that receives an initial high-risk score and is later cleared. The dates below make the timeline and the as-of result explicit:

10 October 2026 — risk 0xdeadbeef 92 chainalysis high mixer exposure

16 October 2026 — transfer approved (score 92 still active)

18 October 2026 — risk 0xdeadbeef 45 internal cleared after review

The history returns both Observations with their sources and timestamps. Both facts remain permanent.

As-of query. An auditor later asks: “What risk score was active as of 16 October 2026 (the date the transfer was approved)?” Cryptomnia returns the Observation that was valid at that moment — score 92 from Chainalysis, asserted on 10 October 2026 — together with its full provenance. The later internal clear (score 45, asserted on 18 October 2026) appears in the full history but is correctly excluded from the as-of result for 16 October. The system does not silently substitute the current score. The as-of query respects the validity interval of each Observation: only the claim that was active at the requested timestamp is returned.

If the Chainalysis source is later judged unreliable for this case, the original 92 score can be marked forgotten. The forgetting event is itself timestamped and queryable, preserving the full chain of institutional knowledge. Any evidence attached to the original observation remains linked and retrievable.

6. Use Cases

Compliance and Regulatory Defense

Organizations can produce a precise, provenance-rich answer to the question regulators and auditors actually ask: what was known at the time the decision was made. Supporting evidence files can be produced alongside the historical score. This reduces the risk of adverse findings driven by incomplete historical records.

Research and Investigations

Investigators can reconstruct what was known at each stage of an inquiry. Re-attributions and new intelligence appear as new Event Objects rather than silent updates, making the evolution of knowledge explicit. Evidence attached at each stage remains available for later review.

Risk Management and Decision Provenance

Product, compliance, and risk teams gain a shared memory of past assessments. This reduces repeated work, supports consistent policy application, and creates a foundation for continuous improvement.

7. Differentiation

Cryptomnia is not another blockchain analytics or transaction-monitoring product. Those systems excel at generating current risk signals. Cryptomnia's role is to make those signals — and the organization's own assessments — historically durable and queryable.

Key distinctions:

- Claim-level temporal fidelity rather than document-level versioning
- Auditable forgetting instead of silent overwrites
- Native as-of queries that respect validity intervals
- Provenance (source, time, confidence) attached to every assertion
- Attachable evidence (files) linked to observations and retrievable later
- Designed to integrate with existing analytics feeds rather than replace them

8. Implementation and Availability

A working Streamlit demonstration stores and retrieves successive risk scores using the Observation/Entity pattern and tag search. As-of queries return the Observation that was valid at the requested timestamp, respecting the validity interval of each claim. Evidence files can be uploaded, linked to an observation, and later downloaded with their original metadata intact. The core memory layer — "what was true when" — is operational.

Cryptomnia is available at cryptomnia.com. A live demonstration can be arranged on request. The system is built for integration with existing risk data pipelines and for use by compliance, audit, and research teams that require defensible historical records.

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9. Conclusion

The permanence of on-chain data is both an opportunity and a liability. Organizations that cannot reconstruct what they knew at the time of a decision face unnecessary regulatory, audit, and operational risk. Cryptomnia supplies the missing memory layer: time-stamped, provenance-rich Event Objects that answer the question that actually matters — what was true when.

By preserving every version, making forgetting auditable, enabling precise historical queries, and retaining the supporting evidence that was available at the time, Cryptomnia turns changing interpretations into durable institutional knowledge.